

Internal Assessment Question Paper – 2

Ramaiah Institute of Technology
(Autonomous Institute, Affiliated to VTU)
Department of CSE

Programme: B.E
Course: Object Oriented Programming
Course Code: CS36
Section: A, B, C
Max Marks: 30

Date: 10/12/2020

Term: Aug to Dec 2020
CIE:1
Sem: 3
Time: 9 to 10 am
Portions: L18 – L35

Instructions to Candidates:

- i. **Question 1 is compulsory** Answer any one full question from 2 or 3.
- ii. Each Question carries 15M.

Sl#	Question	Marks	Bloom's Level	CO
1.	a) i. Recall the clause used when a program does not want to handle exception ii. Class C implements Interface I containing method m1 and m2 declarations. Class C has provided implementation for method m2. Can i create an object of Class C? iii. Write the specifier used for the class, if you do not implement all the methods of an interface while implementing. iv. There are two classes named classA and classB. Both classes are in the same package. Can a private member of classA can be accessed by an object of classB? v. Can we define private and protected modifiers for variables in interfaces?	5	L1	3,4
	b) Write a Java program to find area of a triangle with three sides a, b, c. A triangle can be formed only if $a+b>c$, $b+c>a$, $c+a>b$. First verify whether the above three conditions are satisfied. If any one of them is not satisfied, then throw an exception called Triangle Exception (user defined)	5	L3	3
	c) Explain the mechanism of Event delegation model considering an example of handling ActionEvent.	5	L3	5
2.	a) i. How can you ensure all threads that started from main must end in order in which they started and also main should end in last? ii. What will happen if we don't override Thread class run() method? iii. Which event is generated by clicking a button? iv. Order of execution of constructors in Java Inheritance is a) Base to derived class b) Derived to base class c) Random order d) None v. Which function of Thread class is used to check whether current thread is still running?	5	L2	3,4,5

	b)	List the similarities/difference between an abstract class and interface?	4	L2	3
	c)	Design a java program using threads that shares the synchronized method for multiplication and prints the multiplication table 5 & 6 respectively.	6	L3	4
3.	a)	i. Suppose you have 2 threads (Thread-1 and Thread-2) on same object. Thread-1 is in synchronized method1(), can Thread-2 enter synchronized method2() at same time? ii. What are the thread priority values for Minimum, Maximum & Default? iii. Mention the different ways in which a thread can be created in java. iv. True statement(s) about try catch block a) It is mandatory to have catch block with every try block b) There must be only one catch block followed by try block c) There can be multiple catch block followed by try block. d) All	5	L1	3,4
	b)	Create a base class with an abstract print() method that is overridden in a derived class. The overridden version of the method prints the value of an int variable defined in the derived class. At the point of definition of this variable, give it a nonzero value. In the base-class constructor, call this method. In main(), create an object of the derived type, and then call its print() method. Explain the results.	5	L2	3
	c)	Write java program that i. Creates a Thread object and assigns it to main thread ii. Print the default values of that thread iii. Change name and priority of that thread iv. Print new name and priority separately v. Check and print whether thread is still running or stopped Write the output of above program.	5	L3	4

Course Outcomes meant to be assessed by the IA Test:

CO3:Develop java programs using inheritance, interfaces and packages. (PO-2,3,5 PSO-3)

CO4: Explore the exception handling mechanism and thread synchronization. (PO-2,3,5 PSO-3)

CO5: Design the GUI application using swings and handle the interactions. (PO-2,3,5 PSO3)